

Ten Further Extensions of the Prime-GPF Theory

Mathematical report and Lean formalization status

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15 September 2026

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This report records ten mathematical extensions of the three prime-GPF operations

$$A(p, q) = P^+(p + q + 1), \quad M(p, q) = P^+(pq + 1), \quad E_p(q) = P^+(p^q + 1),$$

with prime inputs and anchors. The mathematical proofs are given below. Formal Lean status is reported separately and is deliberately stricter than mathematical status.

Formalization status

Extension	Mathematical status	Lean status on extensions-5-6-8-9-10-20260915
1	proved below	finite reduction, exact support dimension, and general polylogarithmic bound compiler-verified; sharp coefficient (1) verified when $ T = 1$, including $a = r = 5$; general $ T > 1$ sharp coefficient remains
2	proved below	empty $a = r$ case, support restriction, primitive/nonprimitive decomposition, exceptional-family bounds, exact support dimension, and general polylogarithmic bound compiler-verified; the $1/\zeta(d)$ coefficient (2) remains
3	proved below	exact classification (3), coprimality, residue restriction, exact support $\{3, 5\}$, and quadratic polylogarithmic bound compiler-verified; sharp constant (4) remains
4	proved below	exact classification (5), coprimality, residue restriction, exact support $\{2, 5\}$, and quadratic polylogarithmic bound compiler-verified; sharp constant (6) remains
5	proved below	exponential escape (7), finite visit count, and the literal real-log ₂ form of (8) compiler-verified
6	proved below	finite image bound (9), PNT scaling and limsup bound (10), and deep-image classification (11) compiler-verified
7	proved below	finite restricted-support bound, density zero, prime-relative rate (13), and sharp coefficient (12) when $d_a = 1$ compiler-verified; general $d_a > 1$ sharp coefficient remains
8	proved below	inequalities (14)–(16) and the prime-polynomial no-triple-collision corollary compiler-verified
9	proved below	complete $q = 2$ classification (17) and common output 5 compiler-verified
10	proved below	both algebraic and dynamical separation compiler-verified

The extension development is split into dedicated modules under `PrimeGPF/`; verified leaf modules are imported by the top-level `PrimeGPF.lean`. The general weighted-simplex asymptotic (*) and the Mobius-inversion refinements are the principal remaining analytic infrastructure.

1 Counting lemma

For a finite nonempty set of primes T , put

$$d = |T|, \quad C_T = \frac{1}{d! \prod_{\ell \in T} \log \ell},$$

and let $S_T(Y)$ count positive integers at most Y whose prime factors belong to T . Then

$$S_T(Y) = C_T (\log Y)^d + O_T((1 + \log Y)^{d-1}). \quad (*)$$

Proof. Unique factorization identifies the counted integers with lattice points

$$e_\ell \geq 0, \quad \sum_{\ell \in T} e_\ell \log \ell \leq \log Y.$$

The simplex has volume $C_T (\log Y)^d$. Place a unit cube at each lattice point. Their union is sandwiched between the simplex with bound $\log Y$ and that with bound $\log Y + \sum_{\ell \in T} \log \ell$. The difference of these volumes is $O_T((1 + \log Y)^{d-1})$, proving (*). $\square$

The one-dimensional specialization of (*) needed when $|T| = 1$ is formalized in Lean. The general multidimensional weighted-simplex estimate remains to be formalized.

2 A logarithm saving for every fixed additive fiber

For

$$F_{a,r}^+(x) = \#\{q \leq x : q \text{ prime}, A(a, q) = r\}, \quad r \geq 3,$$

if $r \mid a + 1$, the fiber has at most one input. Otherwise set

$$T = \{\ell \leq r : \ell \text{ prime, } \ell \nmid a + 1\}, \quad d = |T|.$$

Then

$$\limsup_{x \rightarrow \infty} \frac{F_{a,r}^+(x)}{(\log x)^d} \leq C_T. \quad (1)$$

Hence

$$F_{a,r}^+(x) = O_{a,r}((\log x)^{\pi(r)-1}).$$

Proof. If $r \mid a + 1$ and $r \mid a + q + 1$, then $r \mid q$, so primality gives $q = r$. Otherwise take a fiber input $q > r$. Every prime divisor of $a + q + 1$ is at most r , and none divides $a + 1$, since such a divisor would also divide the prime q , contradicting $q > r$. Also $r \mid a + q + 1$. Therefore

$$q \mapsto \frac{a + q + 1}{r}$$

injects these inputs into the T -smooth integers at most $(x + a + 1)/r$. Thus

$$F_{a,r}^+(x) \leq \pi(r) + S_T((x + a + 1)/r),$$

and (*) gives (1). For odd a , $2 \notin T$; for $a = 2$, $3 \notin T$. Hence $d \leq \pi(r) - 1$. $\square$

For example, for $a = r = 5$, $T = \{5\}$, so

$$F_{5,5}^+(x) \leq \frac{\log x}{\log 5} + O(1).$$

The general finite reduction and logarithm-saving exponent are Lean-certified, and the sharp limsup coefficient in (1) is Lean-certified whenever $|T| = 1$, in particular for $a = r = 5$.

3 A $1/\zeta(d)$ improvement for multiplicative fibers

Let

$$F_{a,r}^\times(x) = \#\{q \leq x : q \text{ prime, } M(a, q) = r\}.$$

The fiber is empty if $a = r$. Otherwise put

$$T = \{\ell \leq r : \ell \text{ prime, } \ell \neq a\}, \quad d = |T|.$$

For $d \geq 2$,

$$\limsup_{x \rightarrow \infty} \frac{F_{a,r}^\times(x)}{(\log x)^d} \leq \frac{1}{\zeta(d)d! \prod_{\ell \in T} \log \ell}. \quad (2)$$

Proof. The kernel $aq + 1$ is T -smooth because $a \nmid aq + 1$. For $q > a$, suppose its exponent vector has gcd $g > 1$, so $aq + 1 = u^g$. Then $u - 1 \mid aq$, and $1 \leq u - 1 < q$, since $u \leq \sqrt{aq + 1} < q$. As a, q are distinct primes,

$$u - 1 \in \{1, a\}.$$

Thus every nonprimitive kernel belongs to one of the two families

$$aq + 1 = 2^g \quad \text{or} \quad aq + 1 = (a + 1)^g,$$

which contribute only $O_a(\log x)$ possibilities.

For primitive exponent vectors, Mobius inversion and (*) give, with $L = \log Y$,

$$V_T(L) = \sum_{g \leq L / \min_{\ell \in T} \log \ell} \mu(g)(S_T(e^{L/g}) - 1) = \frac{C_T}{\zeta(d)} L^d + o(L^d).$$

The accumulated error is $O(L \log L)$ for $d = 2$ and $O(L^{d-1})$ for $d \geq 3$. Taking $L = \log(ax + 1)$ proves (2). This is an upper-bound coefficient, not an assertion that the prime fiber attains the coefficient. $\square$

Lean currently formalizes the support restriction, the primitive/nonprimitive dichotomy, the two exceptional perfect-power families, their logarithmic-size bound, the exact support dimension, and the resulting non-sharp polylogarithmic fiber bound. The Mobius-inversion step producing $1/\zeta(d)$ remains.

4 The output-5 multiplicative fiber at anchor 2

For prime q ,

$$M(2, q) = 5$$

if and only if either $q = 2$, or

$$q = \frac{3^\alpha 5^\beta - 1}{2}, \quad \alpha, \beta \geq 1, \quad \alpha \text{ odd}, \quad \gcd(\alpha, \beta) = 1. \quad (3)$$

Every nonexceptional input satisfies $q \equiv 7 \pmod{30}$, and

$$\limsup_{x \rightarrow \infty} \frac{F_{2,5}^\times(x)}{(\log x)^2} \leq \frac{1}{3\zeta(2) \log 3 \log 5}. \quad (4)$$

Proof. For odd q , the odd 5-smooth kernel is

$$2q + 1 = 3^\alpha 5^\beta, \quad \beta \geq 1.$$

If $\alpha = 0$, then $(5^\beta - 1)/2$ is even, so primality forces $q = 2$. Hence $\alpha \geq 1$. Modulo 4, α is odd. If $g = \gcd(\alpha, \beta) > 1$, then g is odd. Putting $t = 3^{\alpha/g} 5^{\beta/g} \geq 15$,

$$q = \frac{t-1}{2}(1+t+\dots+t^{g-1}),$$

contradicting primality. Conversely, (3) directly makes the kernel 5-smooth with greatest prime factor 5. Divisibility by 3 and 5, together with oddness, gives $q \equiv 7 \pmod{30}$.

For (4), the unrestricted weighted exponent triangle has leading coefficient $1/(2 \log 3 \log 5)$. Restricting α to odd values contributes $1/2$, while coprimality over odd common divisors contributes

$$\sum_{g \text{ odd}} \frac{\mu(g)}{g^2} = \frac{4}{3\zeta(2)}.$$

Multiplying gives (4). The formula still requires the resulting q to be prime. □

The exact classification, coprimality condition, congruence restriction, and exact exponent support $\{3, 5\}$ are Lean-certified. The parity-restricted Mobius count giving (4) remains.

5 The output-5 multiplicative fiber at anchor 3

For prime q ,

$$M(3, q) = 5$$

if and only if

$$q = \frac{2^\alpha 5^\beta - 1}{3}, \quad \alpha, \beta \geq 1 \text{ odd}, \quad \gcd(\alpha, \beta) = 1. \quad (5)$$

Every such input satisfies $q \equiv 3 \pmod{10}$, and

$$\limsup_{x \rightarrow \infty} \frac{F_{3,5}^\times(x)}{(\log x)^2} \leq \frac{1}{6\zeta(2) \log 2 \log 5}. \quad (6)$$

Proof. The input $q = 2$ gives output 7. Otherwise the kernel is even, not divisible by 3, and

$$3q + 1 = 2^\alpha 5^\beta, \quad \alpha, \beta \geq 1.$$

Modulo 3 the exponents have the same parity. If both were even, the kernel would be t^2 , and

$$q = \frac{(t-1)(t+1)}{3},$$

with both resulting factors greater than 1, contradicting primality. Hence both exponents are odd. If $g = \gcd(\alpha, \beta) > 1$, put $t = 2^{\alpha/g} 5^{\beta/g}$; then $t \equiv 1 \pmod{3}$ and

$$q = \frac{t-1}{3}(1+t+\dots+t^{g-1}),$$

again contradicting primality. The converse is immediate. Oddness and divisibility of $3q + 1$ by 5 give $q \equiv 3 \pmod{10}$.

For (6), the two odd-exponent restrictions contribute $1/4$, and coprimality contributes $4/(3\zeta(2))$. $\square$

The exact classification, coprimality and parity conditions, congruence restriction, and exact exponent support $\{2, 5\}$ are Lean-certified. The parity-restricted Mobius count giving (6) remains.

6 Explicit exponential escape and sparse orbits

Fix a prime anchor a and set $q_{n+1} = E_a(q_n)$. Except for the constant orbit $a = 2, q_0 = 3$,

$$q_n + 1 \geq 2^n(q_0 + 1). \quad (7)$$

Consequently, for $X \geq q_0$,

$$\#\{n \geq 0 : q_n \leq X\} \leq 1 + \left\lfloor \log_2 \frac{X+1}{q_0+1} \right\rfloor. \quad (8)$$

Proof. The prime-exponent theorem gives $E_a(q) \geq 2q + 1$ for odd prime q , except $(a, q) = (2, 3)$. For $q = 2$, the square-exponent congruence gives $E_a(2) \geq 5 = 2q + 1$. A nonexceptional step therefore outputs at least 5 and cannot subsequently enter the exceptional state 3. Hence

$$q_{n+1} + 1 \geq 2(q_n + 1)$$

at every step. Induction yields (7), and solving (7) for n gives (8). $\square$

The Lean development proves (7), the corresponding finite cardinality bound, and (8) in the literal natural-floor/real-base-two-logarithm form displayed above.

7 Iterated exponential images shrink geometrically

Let

$$I_{a,k} = E_a^k(\mathbb{P}).$$

Then

$$\#(I_{a,k} \cap [2, X]) \leq \pi\left(\frac{X+1}{2^k} - 1\right) + \mathbf{1}_{a=2}. \quad (9)$$

Thus

$$\limsup_{X \rightarrow \infty} \frac{\#(I_{a,k} \cap [2, X])}{\pi(X)} \leq 2^{-k}. \quad (10)$$

Moreover,

$$\bigcap_{k \geq 1} I_{a,k} = \begin{cases} \{3\}, & a = 2, \\ \emptyset, & a \neq 2. \end{cases} \quad (11)$$

Proof. If a nonexceptional starting prime has its k -th iterate at most X , (7) gives

$$q_0 \leq \frac{X+1}{2^k} - 1.$$

Counting possible starting primes bounds the number of distinct outputs; the exceptional fixed orbit contributes at most one extra value. This proves (9). The prime number theorem then gives (10).

If a nonexceptional prime r has a k -step ancestor, then

$$r + 1 \geq 2^k(q_0 + 1) \geq 3 \cdot 2^k,$$

so its ancestry depth is finite. Only $E_2(3) = 3$ has arbitrarily deep ancestry, proving (11). $\square$

The Lean development proves (9), the PNT fixed-scale ratio and limsup estimate (10), and the exact all-depth classification (11).

8 Quantitative rarity of additive-multiplicative collisions

Fix a prime a , put

$$B_a = a^2 + a - 1, \quad R_a = P^+(B_a),$$

and

$$T_a = \{\ell \leq R_a : \ell \text{ prime, } \ell \nmid a+1\}, \quad d_a = |T_a|.$$

Then

$$D_a(X) = \#\{q \leq X : q \text{ prime, } A(a, q) = M(a, q)\}$$

satisfies

$$D_a(X) \leq C_{T_a} (\log X)^{d_a} + O_a((\log X)^{d_a-1}), \quad (12)$$

and therefore

$$\frac{D_a(X)}{\pi(X)} = O_a\left(\frac{(\log X)^{d_a+1}}{X}\right). \quad (13)$$

Proof. The polynomial-divisor restriction says every common output r divides B_a , so $r \leq R_a$ and the additive kernel is R_a -smooth. For $q > R_a$, the same exclusion argument as in extension 1 shows that the kernel uses only primes in T_a . Thus

$$D_a(X) \leq \pi(R_a) + S_{T_a}(X + a + 1).$$

Apply (*) for (12) and the prime number theorem for (13). The set T_a is nonempty because $B_a \equiv -1 \pmod{a+1}$, so its greatest prime divisor cannot divide $a+1$. $\square$

Lean proves the finite restricted-support reduction, a degree- d_a polylogarithmic upper bound, density zero, and the prime-relative Big-O rate (13). The sharp coefficient in (12) is additionally formalized when $d_a = 1$; the general $d_a > 1$ coefficient remains tied to (*).

9 Additive-exponential collisions are confined to a finite region

If

$$A(p, q) = E_p(q),$$

then

$$q \leq p, \quad (14)$$

except at $(p, q) = (2, 3)$. If p, q are odd and $p + q + 1$ is composite, then

$$p \geq 5q + 2. \quad (15)$$

For a triple collision,

$$q \leq \min\left(p, \frac{P^+(p^2 + p - 1) - 1}{2}\right). \quad (16)$$

Proof. Outside the exceptional pair, a common output r satisfies

$$2q + 1 \leq r \leq p + q + 1,$$

which gives (14). If the odd additive kernel $N = p + q + 1$ is composite, then its cofactor after division by $r = P^+(N)$ is an odd integer at least 3. Therefore

$$p + q + 1 = N \geq 3r \geq 6q + 3,$$

which is (15). For a triple collision, the additive-multiplicative polynomial restriction gives $r \mid p^2 + p - 1$, so

$$2q + 1 \leq r \leq P^+(p^2 + p - 1),$$

and (16) follows together with (14). The exceptional pair $(2, 3)$ is not triple because its multiplicative output is 7 while the additive and exponential outputs are 3. $\square$

If $p \geq 3$ and $p^2 + p - 1$ is prime, no triple collision can occur at anchor p : the divisor restriction would force the common output to be $p^2 + p - 1$, whereas (14) bounds it by $p + q + 1 \leq 2p + 1 < p^2 + p - 1$.

All three inequalities and this prime-polynomial corollary are Lean-certified.

10 Complete classification of triple collisions with $q = 2$

For prime p ,

$$A(p, 2) = M(p, 2) = E_p(2) \iff p \in \{2, 7\}. \quad (17)$$

The common output is 5.

Proof. The existing second-input-2 collision theorem forces any common output to be 5. Direct calculation gives the two solutions: for $p = 2$, the three kernels are all 5; for $p = 7$, the kernels are 10, 15, and 50, all with greatest prime factor 5.

Conversely assume $p \neq 2$. From $M(2, p) = 5$ one obtains $p \equiv 1 \pmod{3}$. Since p is odd, $p^2 + 1$ contains exactly one factor 2 and no factor 3. Thus

$$p^2 + 1 = 2 \cdot 5^k, \quad k \geq 1.$$

Modulo 3 forces k even, hence $k \geq 2$. Modulo 16 then gives $p^2 \equiv 1 \pmod{16}$.

The additive condition gives

$$p + 3 = 2^\alpha 5^\beta, \quad \alpha, \beta \geq 1,$$

because 3 does not divide $p + 3$. If $\beta \geq 2$, then $p \equiv -3 \pmod{25}$, so $p^2 + 1 \equiv 10 \pmod{25}$, contradicting $k \geq 2$. Hence $\beta = 1$.

Modulo 3, $5 \cdot 2^\alpha = p + 3 \equiv 1$, so α is odd. Since $p^2 \equiv 1 \pmod{16}$,

$$p \equiv 1, 7, 9, 15 \pmod{16}.$$

In all four cases the 2-adic valuation of $p + 3$ is at most 2. Hence $\alpha \leq 2$, and oddness forces $\alpha = 1$. Therefore $p + 3 = 10$, so $p = 7$. $\square$

The equivalence (17) and the common-output-5 conclusion are Lean-certified.

11 Structural separation from the homogeneous additive GPF family

Define

$$H(p, q) = P^+(p + q).$$

11.1 Algebraic separation

There is no magma homomorphism from H into any of A, M, E , and no injective magma homomorphism from any of A, M, E into H .

Proof. For every prime p ,

$$H(p, p) = P^+(2p) = p,$$

so every element of H is idempotent. None of A, M, E has an idempotent because p divides none of

$$2p + 1, \quad p^2 + 1, \quad p^p + 1.$$

A homomorphism preserves idempotents, which rules out homomorphisms from H to any of the three new magmas. An injective homomorphism into an idempotent magma would force the source operation to be idempotent, ruling out injective homomorphisms in the reverse direction. $\square$

This algebraic statement is Lean-certified on the branch.

11.2 Dynamical separation

For odd prime anchors a, c , every orbit of

$$H_c(q) = P^+(c + q)$$

is eventually periodic, while every orbit of E_a escapes exponentially. More strongly:

- no finite-to-one map $h : \mathbb{P} \rightarrow \mathbb{P}$ satisfies $h \circ E_a = H_c \circ h$;
- no map at all satisfies $h \circ H_c = E_a \circ h$.

Proof. If q is odd, $c + q$ is even and composite, so

$$H_c(q) \leq \frac{c + q}{2}.$$

For $q = 2$, trivially $H_c(2) \leq c + 2$. Thus every input above $c + 2$ strictly decreases, while the finite set of primes at most $c + 2$ is forward invariant. Every orbit therefore eventually enters that finite set and is eventually periodic.

An escaping E_a -orbit contains infinitely many distinct states. A finite-to-one intertwining map cannot send all of them into one finite H_c -orbit. Conversely, $H_c(c) = c$; an intertwining map in the reverse direction would send this fixed point to an E_a -fixed point, but no such fixed point exists for odd a . $\square$

The eventual-periodicity result and both intertwining obstructions are Lean-certified in `PrimeGPF/Extension 10Dynamics.lean`.

12 Remaining directions

The results above do *not* establish asymptotic equivalences for the actual prime fibers, nor their infinitude, nor a global classification of triple collisions with odd second input. The principal remaining formalization task for extensions 1–4 and 7 is the general weighted-simplex counting estimate (*); the ζ -refinements in extensions 2–4 additionally require Mobius inversion, including parity-restricted primitive-vector counts for (4) and (6). The one-dimensional sharp cases of (1) and (12), the full PNT passage in (10), and the prime-relative rate (13) are already formalized.